ADITYA UNIVERSITY

(Formerly Aditya Engineering College (A))

PROBABILITY& STATISTICS

Assignment-II

1. A population consists of five numbers 3, 6, 9, 15 and 27. Consider all possible samples of size 3 that can be drawn without replacement from this population. Find i) The population mean ii) The population standard deviation ii) The mean of the sampling distribution of the means iv) The standard deviation of the sampling distribution of the means.
2. A population consists of the numbers 2, 3, 6, 8 and 11. Consider all possible samples of size 2 that can be drawn with replacement from this population. Find i) The population mean ii) The population standard deviation iii) The mean of the sampling distribution of the means iv) The standard deviation of the sampling distribution of the means.
3. Find the median for the following table

Group	50-54	55-59	60-64	65-69	70-74	75-79
Frequency	4	8	10	9	7	6

4. What are the types of data? Explain the methods of collection of data.
5. A random sample of size 100 is taken from an infinite population having the mean μ=76 and the variance $\sigma^2 = 256$. Find the probability that will be i) lies between 75 and 78 ii) More than 72.
6. Find 95% confidence limits for the mean of a normally distributed population from which the following sample was taken 15, 17, 10, 18, 16, 9, 7, 11, 13, 14, 20, 13, 19, 25, 18, 16.
7. Two researchers adopted different sampling techniques while investigating the same group of students to find the number of students falling in different intelligence levels. The results are as follows:

Researcher	No. of students in each level				Total
	Below Average	Average	Above Average	Genius	
X	86	60	44	10	200
Y	40	33	25	2	100
Total	126	93	69	12	300

Would you say that the sampling techniques adopted by the two researchers are significantly different?

8. The mean yield of wheat from a district was 210 pounds with S.D 10 pounds per acre from a sample of 100 plots. In another district the mean yield was 220 pounds with S.D 12 pounds from a sample of 50 plots. Test whether there is any significant difference between the mean yields of crops in the two districts.

9. Ten soldiers participated in a shooting competition in the first week. After intensive training they participated in the competition in the second week. Their scores before and after training are given as follows:

Scores Before	67	24	57	55	63	54	56	68	33	43
Scores after	70	38	58	58	56	67	68	75	42	38

Do the data indicate that the soldiers have been benefited by the training?

10. A Sample of 64 students have a mean weight of 70kgs. Can this be regarded as a sample from a population with mean weight 56kgs and standard deviation 25kgs?

11. Fit a parabola to the following data

x	0	1	2	3	4
y	1	1.8	3.3	4.5	6.3

12. For the following data, fit the curve of the form $y = ae^{bx}$.

x	55	56	58	59	60	60	62
y	35	38	38	39	44	43	45

13. Calculate Rank correlation for the following data

x	48	33	40	9	16	16	65	24	16	57
y	13	13	24	6	15	4	20	9	6	19

14. Calculate Karl Pearson's coefficient of correlation for the following data

x	12	9	8	10	11	13	7
y	14	8	6	9	11	12	3

15. The equations of two regression lines are $7x - 16y + 9 = 0$ and $5y - 4x - 3 = 0$, find the coefficient of correlation and the means of x and y.

16. Ten persons are assessed by three judges in a beauty contest. Determine the closeness between opinions of judges by rank correlation coefficient.

Score assigned by J-1	65	70	72	76	77	79	72	80	77	72
Score assigned by J-2	59	71	77	71	69	79	69	65	67	80